GOLDSCHMIDT 2026

Orbitrap-IRMS Abstracts (Posters and Oral Presentations)

MONDAY

<u>Oral Presentations</u>

Gontijo Machado et al., “From Transient to Isotope Ratio: Targeted Signal Processing Strategies for Compound-Specific Isotope Analysis by Orbitrap MS”

High-resolution Orbitrap mass spectrometry is emerging as a powerful platform for compound-specific stable isotope analysis of intact molecules. However, achieving the precision required for isotopic applications remains challenging in complex natural matrices. In Fourier transform mass analyzers, ion motion is detected as a time-domain transient (FID) and converted to mass spectra via frequency estimation and calibration [1,2]. Because image-current frequency is sensitive to ion population, localized space-charge interactions induce small but systematic frequency shifts. In conventional high-resolution omics applications, sub-ppm shifts typically introduce negligible quantitative bias. In isotope ratio measurements, however, even a ~1% relative intensity distortion between isotopologues can propagate to errors on the order of 10‰, rendering standard optimized “black-box” processing insufficient for high-precision isotope applications.

Here we directly access and reprocess raw Orbitrap transients to systematically evaluate how signal processing and calibration strategies affect isotopologue ratio stability. Organic acids were analyzed in synthetic standards and complex produced water matrices by ESI-Orbitrap-MS. We examine how FID-to-spectrum conversion (including frequency estimation, peak definition, centroid determination, and calibration mapping) interacts with localized space-charge-induced frequency shifts to influence measured M0/M+1 intensity ratios. Particular emphasis is placed on calibration strategy: conventional broadband calibration is compared to targeted narrow-window approaches, including lock-mass correction and two-point calibration anchored exclusively to the monoisotopic (M0) and heavy (M+1) isotopologues. Constraining calibration to the isotopologue pair of interest reduces local frequency-to-mass mapping asymmetry, thereby mitigating differential centroid displacement between M0 and M+1 peaks under varying ion population regimes. This framework enables clearer discrimination between instrumental artifacts and true matrix-driven isotope effects

Traditional IRMS achieves high precision through analyte-specific tuning of simple combustion products (e.g., CO_2, N_2). In contrast, Orbitrap instruments are designed for multiplexed, untargeted analysis of hundreds to thousands of species. We propose that high-precision isotope ratio applications similarly require analyte-directed signal processing strategies. Establishing targeted FID reprocessing workflows represents an essential step toward robust Orbitrap-based compound-specific isotope analysis in complex geochemical systems.

References: [1] A.Makarov, Electrostatic axially harmonic orbital trapping: A high-performance technique of mass analysis, Anal. Chem. **72**, 1156 (2000); [2] M.Scigelova and A.Makarov, *Orbitrap Mass Analyzer - Overview and Applications in Proteomics*, in *Proteomics* (2006).

Poster Presentations

None noted.

TUESDAY

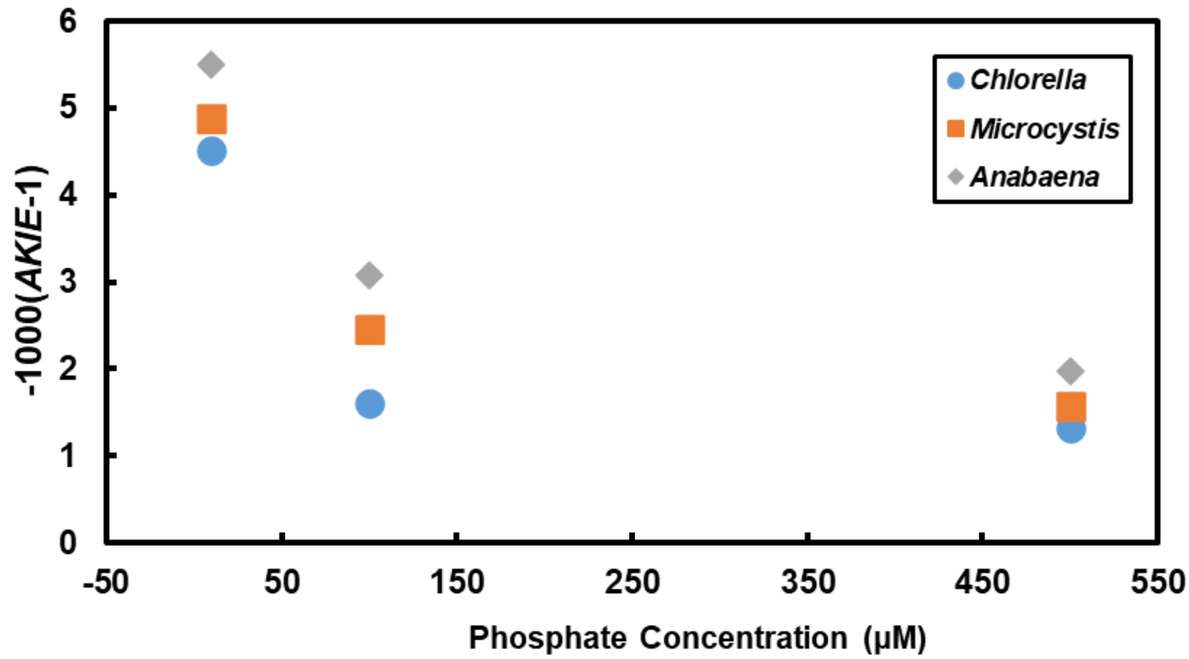
Oral Presentations

Wei et al., "Larger oxygen isotope effects during phosphate uptake in lower-phosphorus water"

Isotope fractionation during microbial oxyanion reduction (e.g., sulfate, nitrate, and perchlorate) is overwhelmingly dominated by intracellular enzymatic bond-breaking reactions, rendering isotope effects associated with membrane transport largely obscured. In contrast, phosphate (PO_4^{3-}) uptake involves no redox transformation or bond cleavage during translocation across the cell membrane, making it uniquely suited for isolating isotope effects arising solely from transporter binding and translocation. Our recent phosphate oxygen isotope ($\delta^{18}O_P$) data from extremely low-[PO_4^{3-}] lakes (at detection limit of ~0.1 μM) revealed that the apparent kinetic isotope effect (*AKIE*) (1000(*AKIE*-1) ~ -6‰) is nearly twice the previously determined value of ~ -3‰ starting at [PO_4^{3-}] of 200 μM, suggesting the oxygen isotope effect associated with PO_4^{3-} uptake at low-[PO_4^{3-}] conditions could be even larger.

Here we test the hypothesis that the transport *AKIE* associated with PO_4^{3-} uptake through cell membrane scales smoothly with ambient PO_4^{3-} concentration. Micro-molar [PO_4^{3-}] $\delta^{18}O_P$ measurement are now possible, thanks to our recently developed ESI-Orbitrap-MS approach [1,2]. We have conducted a set of culture experiments using a green alga (*Chlorella*) and two cyanobacteria (*Microcystis* and *Anabaena*) under initial PO_4^{3-} conditions of 10, 100, and 500 μM. Residual PO_4^{3-} samples were taken at the respective exponential growth stages. Results show that the 1000(*AKIE*-1) values for *Chlorella, Microcystis* and *Anabaena* change from -1.6‰, -2.5‰ and -3.1‰ to -4.6‰, -4.9‰, and -5.5‰, respectively as the initial experimental PO_4^{3-} concentration decreases from 500 to 100, and to 10 μM (Fig. 1), confirming a higher degree of normal *AKIE* under lower PO_4^{3-} concentrations. Experiments with additional initial PO_4^{3-} concentrations are required to confirm if indeed low-affinity and high-affinity transporters have different fundamental fractionation factors or the observed trend is merely the result of different degrees of backward PO_4^{3-} flux among the different experimental conditions.

References: [1] Wang Z, Hattori S, Peng Y, Zhu L, Wei Z, Bao H. *Analytical Chemistry* 2024, **96**(11)**:** 4369-4376; [2] Wei Z, Wang B, Zhu L, Hong Y, Wang Z, Yan H*, et al. Applied Geochemistry* 2025, **194:** 106610.



Poster Presentations

Yang & Xie, "High-Precision Position-Specific Hydrogen Isotope Analysis of Fatty Acids via GC-Orbitrap"

Stable isotopes have become indispensable tools for deciphering complex processes in both environmental and biological systems. In biological system, isotope fractionation during metabolism offers a unique window into pathophysiology. However, conventional GC-IRMS methods are limited to molecular-average isotopic values, obscuring critical position-specific variations (δ^2H) that are highly sensitive to metabolic shifts. In this study, we developed a high-precision method for position-specific hydrogen isotope analysis of fatty acids using a custom-modified GC-Orbitrap mass spectrometer. By leveraging the high mass resolution of the Orbitrap, we achieved direct characterization of intramolecular hydrogen distributions without complex chemical degradation. Our method demonstrates exceptional sensitivity, requiring only 7.6 nmol of sample to reach experimental precisions of 5.4 ‰ for $C_4H_7O_2^+$ and 8.8 ‰ for $C_5H_9O_2^+$ (1 SD, n = 5). Validation using labelled palmitic acids conformed negligible hydrogen scrambling and a strong correlation (R^2 = 0.991, slope

= 1.044) with whole-molecule GC-IRMS measurements across a wide composition range (-223 – 418 ‰). By resolving position-specific isotopic fingerprints, the research provides a robust tool for tracing cellular metabolic condition, offering new avenues for the early diagnosis and monitoring of metabolic disorders and cancer through naturally occurring isotopic signatures.

WEDNESDAY

Oral Presentations

None noted.

Poster Presentations

None noted.

THURSDAY

Oral Presentations

Eiler, "Molecular Isotopic Structure as a Tool for Study of Earth-Surface Materials"

Stable isotopes can trace critical elements from sources to sinks in biogeochemical cycles; establish biogeochemical mechanisms affecting compounds of interest; quantify environmental temperatures; and indirectly constrain humidity, oxidation state, moisture budget or other environmental properties. But this set of methods often struggles with the inability to deconvolve multiple drivers of isotopic variations without prior assumptions.

Advances in technologies and measurement concepts, mostly over the last 25 years, have dramatically improved our ability to address such problems. 'Mass laws' for O and S isotopes have narrowly but powerfully impacted studies of the water cycle, atmospheric photochemistry, and S metabolisms. Potentially broader capabilities are provided by measurements of isotopic structures of molecules — i.e., heterogeneous distributions of isotopes across non-equivalent atomic positions ('site specificity'), and/or the propensities for rare isotopes to combine in the same molecule in amounts different from statistical probabilities ('isotopic clumping'). Molecular isotopic structures of CO_2, N_2O, CH_4, O_2, N_2, and H_2 are now familiar (100+ peer review papers per year) ways of measuring paleotemperature (i.e., carbonate clumped isotope thermometry) and studying cycling of small, biogeochemically active gases.

Perhaps the newest and most complex technical advance is high-dimensionality isotopic fingerprinting of biochemical compounds and pollutants. It is now possible to measure ~5-100 independent statements of isotopic structure simultaneously on a wide range of molecules. These data enable forensic assignment with unprecedented specificity and, when combined with machine-learning classifiers, exceptional sensitivity. Such measurements also enable a new, sophisticated means of studying metabolism and its environmental dependencies. Recent and emerging works prove the value of this approach in studies of sugars, organic acids, amino acids, steroidal compounds, and vanillin and other methoxy-bearing biomolecules. Further expansion of these measurements and, critically, their interpretations through advances in understanding and quantitative modeling of metabolic isotope effects will give rise to a vigorous new sub-discipline of stable isotope geochemistry.

Moran et al., "Enabling stable isotope fingerprinting of PFAS using Orbitrap IRMS: A tool for source attribution of fluorinated organic compounds"

Per- and polyfluoroalkyl substances (PFAS) are a critical threat to ecosystem and human health. PFAS exhibit a known and growing list of toxic effects, making their mitigation a priority for limiting negative environmental consequences. Engineered solutions for deconstructing PFAS are in development and may become a powerful tool for environmental remediation. However, the prevalence of various PFAS compounds in a wide range of industrial sites, agricultural systems, and consumer products can make it challenging to link an environmental contaminant to a specific source – an important connection for guiding and improving the efficiency of remediation efforts. Stable isotope fingerprinting has demonstrated utility for linking samples to their source in contaminant studies, chemical forensics, and food authentication amongst other efforts. Isotope ratio mass spectrometry (IRMS) is the workhorse for stable isotope fingerprinting and has been the dominant analytical platform for previous contaminant studies. However, IRMS analysis requires complete analyte conversion to simple gases (which can be challenging given the chemical recalcitrance of PFAS), necessitates relatively large sample sizes (which is incompatible for compounds like PFAS that are toxic at trace concentrations), and cannot resolve distinct, intra-molecular stable isotope signatures (that could provide an additional forensic signature). To circumvent these challenges, we sought to evaluate whether Orbitrap IRMS could enable stable isotope fingerprinting of PFAS. We used perfluorooctanoic acid (PFOA) as a model PFAS compound for initial testing and developed an Orbitrap method for measuring $\delta^{13}C$ of both the full molecule and a decarboxylated fragment of PFOA. Of the 23 commercial samples we examined, 91% of pairwise comparisons confirmed distinct isotope values, suggesting sufficient variation exists to serve as a fingerprinting tool. Further, intramolecular stable isotope distribution within PFOA showed correlation with structural isomers of the compounds which may provide an additional signature indicative of synthetic method. While the true capabilities of Orbitrap analysis for molecular-level stable isotope analysis are still coming into focus, Orbitrap has transformative potential for unlocking stable isotope signatures in PFAS to help combat these emerging contaminants.

Arevalo et al., "Laser-Based Mass Spectrometry for the Chemical Analysis of Precious Planetary Samples"

Laser-based analytical techniques, such as laser desorption mass spectrometry (LDMS) and laser ablation inductively coupled plasma mass spectrometry (LA-ICPMS), enable the untargeted characterization of solid sample composition ([1] and references therein). The spatial resolution of such methods ranges from the nm- (e.g., ablation depth per laser shot) to the μm- (e.g., typical focused beam diameters) to the mm-scale (e.g., 2D chemical imaging via active beam scanning or sample manipulation). Collocated observations of mineralogy, elemental/isotopic composition, and organic molecule inventory enable insights into sourcing/provenance, reconstructions of local environmental conditions, and geological context for putative biosignatures. Laser-based mass spectrometry techniques are thus well adapted for the comprehensive chemical analysis of precious planetary materials, as supported by the inclusion of instrumentation on several upcoming planetary missions (e.g., ExoMars/MOMA; [2]) as well as modern-day workflows for ground-based investigations into asteroidal return samples (e.g., Bennu [3]).

Miniaturized, field-deployable hardware facilitates in situ investigations by enabling transportation of analytical instrumentation to the curation facility rather than requiring that the samples be shipped to/from multiple organizations, reducing the risk for contamination from additional handling requirements and/or disparate institutional capabilities/processes. Miniaturized laser-based instruments are particularly valuable when considering the containment protocols and biosafety measures associated with returning planetary materials that could harbor extant life [4], such as subsurface samples from Mars. Miniaturized instruments could be installed semi-permanently into Biosafety Level 4 spaces, like the Mars Sample-Return Receiving Facility (MSRRF), circumventing the risk of backwards contamination without needing to sterilize (and potentially alter) the samples.

In the M-CLASS Laboratory, we have developed several low-mass (< 20 kg), low-power (< 100 W peak) laser-based instruments that can be deployed in the field or relocated to curation facilities. Here, we demonstrate the value of: deriving 2D chemical images to map sample heterogeneities; quantifying elemental abundances to establish lithology and track geochemical proxies; determining mineral modal proportions; and, characterizing the diversity, distribution, and molecular complexity of insoluble organic matter.

References: [1] Farcy and Arevalo (2025) *Treatise on Geochemistry (3rd Edition)*; [2] Goesmann et al. (2017) *Astrobiology*; [3] Glavin et al. (2025) *Nature Astronomy*; [4] Teece et al. (2025) *Astrobiology*.

Hattori et al., "Tracing atmospheric DMS chemistry using the triple oxygen isotope of MSA"

The atmospheric sulfur cycle exerts a central control on Earth's radiative balance by regulating aerosol formation, cloud microphysics, and biogeochemical feedbacks. Within this cycle, oxidation of marine dimethyl sulfide (DMS) represents the dominant natural source of non-sea-salt sulfate and methanesulfonate (MSA), yet the partitioning of sulfur among gas-phase MSA, aqueous MSA, sulfuric acid, and particulate sulfate remains a major uncertainty in climate models. Because these products contribute differently to new particle formation, condensational growth, and cloud condensation nuclei (CCN) abundance, even modest errors in DMS oxidation branching can propagate into substantial biases in radiative forcing. However, the branching mechanisms of DMS oxidation and their climatic implications remain poorly constrained. Conventional concentration-based metrics such as the MSA/SO_4^{2-} ratio are affected by uncertainties in both DMS oxidation chemistry and sulfate source strengths, limiting their ability to uniquely constrain oxidation pathways.

Independent insight into DMS oxidation chemistry can be obtained from the mass-independent oxygen isotopic composition ($\Delta^{17}O = \delta^{17}O - 0.52 \times \delta^{18}O$) of MSA. These measurements were enabled by the recent development of an electrospray ionization Orbitrap mass spectrometric method that allows determination of $\Delta^{17}O$ values in intact MSA molecules (Hong et al., 2025 Anal. Chem.). Here we present direct observations of the $\Delta^{17}O$(MSA) values in Antarctic aerosols collected at Syowa Station during 2020–2021. Observed $\Delta^{17}O$(MSA) values exhibit pronounced seasonal variability, with lower values in summer and higher values in winter, indicating seasonally varying oxidation pathways in the Antarctic atmosphere. Comparison with simulations from the GEOS-Chem chemical transport model shows that the default chemical mechanism for DMS chemistry systematically overestimates $\Delta^{17}O$(MSA) values despite reproducing the observed MSA/SO_4^{2-} ratio. Sensitivity experiments indicate that lowering $\Delta^{17}O$(MSA) endmember from the gas-phase MSA production and/or reducing contributions from BrO-involved reactions substantially improve agreement with observations. We will also discuss future directions in which isotope-constrained oxidation chemistry is incorporated into aerosol microphysics and radiative forcing calculations to help close the sulfur budget and reduce uncertainties in climate feedbacks associated with marine sulfur emissions.

References: [1] Hong et al., (2025) Triple Oxygen Isotope Analysis of Methanesulfonate Using the SO_3^- Fragment in ESI-Orbitrap-MS, *Analytical Chemistry* 97, 27, 14339–14348.

Poster Presentations

Zhu et al., "Isotope compositions of diverse natural perchlorate measured by ESI-Orbitrap-MS"

Multiple mechanisms have been proposed to account for the curiously rich ClO_4^- in Martian soils. The stable isotopes ($\delta^{37}Cl$, $\delta^{18}O$, and $\Delta^{17}O$) of ClO_4^- provide fingerprints for its formation pathways as well as post-depositional processes. Yet similar isotopic tools applied to terrestrial perchlorate reveal a striking discrepancy: existing reaction-transport models [1] consistently overestimate $\Delta^{17}O$ values compared to the few observed ones, suggesting either an overestimation of ozone-mediated oxidation in models or significant contributions from non-ozone pathways in natural samples. Contribution from non-ozone oxidation pathways should have a $\Delta^{17}O$ value at ~0‰, whereas contribution from ozone-mediated pathways should yield $\Delta^{17}O$ >20‰ in natural samples (e.g., snow, rain, aerosols).

To test the prediction, we collected ClO_4^- from anthropogenic sources, Yangtze River water, and East China snow, as well as previously collected Atacama Desert caliche. Isotopic measurements of natural ClO_4^- usually require the processing of a large quantity of host water or rock samples with complex matrices, which has been analytically prohibitive until now. We have developed an ESI-Orbitrap-MS method capable of high-precision isotopic analysis of <10 nmol ClO_4^- with an accuracy better than 1‰ for all $\delta^{18}O$, $\delta^{37}Cl$, and $\Delta^{17}O$ parameters [2]. Natural samples went through a resin-based enrichment and Ion Chromatography (IC) fraction collection and purification protocol. Preliminary results show that Yangtze River water and East China snow have a large fraction of ClO_4^- that has no ^{17}O anomaly. The accompanied $\delta^{18}O$ and $\delta^{37}Cl$ data range from −19‰ to −9‰ and 0‰ to 5‰, respectively. Our expanding natural ClO_4^- $\delta^{18}O$, $\delta^{37}Cl$, and $\Delta^{17}O$ database of high spatial/temporal resolution promises the eventual resolution of the discrepancies between models and observation.

References: [1] Chan, Jaeglé, Campuzano-Jost, Catling, Cole-Dai, Furdui, Jackson, Jimenez, Kim, Wedum & Alexander (2023), *Geophys. Res. Lett.* 50, e2023GL102745; [2] Zhu, Hong, Hattori, Wang, Wei, Peng, Kuhlbusch & Bao (2026), *Anal. Chem.*

Briggs et al., “Development of analytical method for clumped isotope measurements in soil nitrate”

Stable isotope analysis represents an important indirect measurement for determining the fate of nitrogen in soils, as stable isotope ratios provide a record of past metabolic transformations a sample has undergone. Therefore, stable isotope ratios can be used to distinguish between two major nitrate loss pathways found in soils: microbial denitrification and leaching to groundwater [1]. However, traditional analytical methods for measuring ^{15}N:^{14}N ratios in nitrate samples produce high isotopic variance at low environmental concentrations, and requires assumptions on the initial nitrate isotope ratios, making this approach uncertain for estimating nitrate losses via these individual pathways [2]. Additionally, to avoid the loss of relevant molecular isotopic (isotopologue) information through the sample conversion process that is necessary with traditional IRMS, electrospray-ionization Orbitrap mass spectrometry has recently been successful in measuring isotopes, including clumped isotopes, of intact nitrate molecules with high precision [3]. With adequate analytical methods for measuring nitrate clumped isotopes now a possibility, isotopologue fractionation occurring during microbial nitrate production (nitrification) and consumption (denitrification) can be quantified to assist in tracing nitrates through soils. This research project aims to develop an analytical method for soil nitrate isotopologue analysis with an emphasis on clumped isotopes using ESI-Orbitrap mass spectrometry. This method can therefore be used to determine the isotopologue fractionation of microbial nitrification and denitrification to assist in distinguishing these nitrate transformation pathways in soils. If successful, clumped isotope analysis will allow for differentiating between source and sink processes of nitrate in soils.

References: [1] Bachmann et al. (2026), *Geoderma, 467, 117746*; [2] Biasi et al. (2022), *Rapid Communications in Mass Spectrometry, 36*(22), e9370; [3] Hilkert et al. (2021), *Analytical Chemistry, 93*(26), 9139-9148.

FRIDAY

<u>Oral Presentations</u>

Eiler, "The Vast Unexplored Landscape of Nature's Isotopic Structure - *V.M. Goldschmidt Award Lecture*"

Isotope geochemistry is an outgrowth of discoveries in nuclear and chemical physics and inventions for analytical chemistry. The field generally tries to mold those discoveries and inventions into a toolkit for answering questions in the earth, environmental and space sciences. However, there is also a role for isotope geochemistry to turn around and look in the other direction, seeking new frontiers in fundamental chemistry, physics and technology motivated by questions posed by the natural world.

Among the largest unresolved questions in natural chemistry concerns the properties and distributions of isotopic forms of molecules. This subject might seem to have been solved by our academic great, great grand advisors, but there are some details still to be settled: There exists a total of est.1020 unique isotopologues — isotopic versions of molecules — summed across all natural and common synthetic compounds. Up until a few years ago, an order of a hundred had been analyzed directly in natural materials at useful precision and sensitivity. However, the last several years of developments in isotope ratio mass spectrometry have made possible precise measurements of ~1011-12 of these species at natural isotope abundances and modest sample sizes. I.e., our capacity to explore the isotopic diversity of nature very recently expanded by something like a factor of a billion. How do these newly analyzable species behave? What controls their distributions? What new geochemical tools will emerge from their systematic study? This subject seems overwhelmingly complex, but might that complexity be turned to our advantage by way of the equally new tools of machine-learning-informed data science?

I will explore these questions by posing first-order problems in the origins, forms, distributions and effects of volatile elements in the early solar system, from nebular and disk history to prebiotic chemistry and detections of life. I will then detail how these problems motivated invention of new analytical technologies and exploration of chemical physics in new contexts, letting us use nature's vast landscape of molecular isotopic structures to address some of its most challenging problems.

Monticelli et al., "ESI-Orbitrap Procedure for the Reliable Determination of Lead Isotopic Ratios in Sediment Samples"

We recently developed a novel strategy to convert a conventional Orbitrap™ mass spectrometer into a platform for metal isotope ratio analysis, requiring no hardware modifications and enabling rapid, interference-free measurements. The method relies on the complexation of metal ions with EDTA to enhance ionization efficiency, followed by quadrupole filtering to isolate the precursor ions. Subsequent collision-induced dissociation releases free metal ions, enabling direct ultra-high-resolution isotopic analysis [1].

Here we will demonstrate for the first time that this procedure provides accurate and precise values for lead isotopic ratios in samples of sediment cores collected in two different subbasins of Lake Como (Italy). The optimization strategy enabled full adaptation of the method to the specific characteristics of the samples. Samples were first diluted to maintain matrix elements and lead concentrations within acceptable limits, while appropriate EDTA concentrations ensured complete lead complexation. Mass bias between samples and standards was minimized by exploiting a distinctive capability of Orbitrap instrumentation: the injection of a fixed number of ions into the analyzer. By tuning the instrumental parameters, identical ion populations were introduced for all analyzed solutions, drastically reducing matrix-dependent space charge effects, ensuring highly consistent conditions between samples and standards. Several mass bias correction schemes were then assessed, including standard sample bracketing, internal standard and the optimized regression method (ORM). The simple bracketing method emerged as the optimal approach, delivering results comparable to or superior to the other methods, while significantly simplifying data treatment.

The procedure was validated through the analysis of 31 sediment samples using both the optimized method and Multi Collector ICP-MS, with no systematic deviations observed, thereby confirming the robustness and accuracy of the ESI-Orbitrap approach. Future perspectives will be also briefly outlined, including expansion of the range of analyzable elements and strategies to enhance analytical performances.

Reference: [1] G. Roncoroni, D. Spanu, G. Binda, D. Monticelli, Anal. Chem. 2025, 97, 25, 13176–13183, https://doi.org/10.1021/acs.analchem.5c01033.

He et al., "Nitrogen isotopes of amino acids in meteorites and fossils"

Nitrogen isotope ratios ($\delta^{15}N$) of amino acids constrain nitrogen sources and metabolic processes, enabling applications in ecology and organismal biology, such as establishing trophic position [1] and reconstruct feeding behavior in extant and extinct animals [2]. Applications to fossil materials, however, remain limited because proteins commonly targeted for such analyses, particularly collagen, degrade rapidly (<100 ka). Dental enamel is a rare exception, as it can preserve endogenous proteins over deep time, offering unique potential for dietary reconstruction. Yet the low protein content of enamel (~0.5–2 wt%) has historically precluded amino acid–specific nitrogen isotope analysis, because conventional gas chromatography–isotope ratio mass spectrometry (GC-IRMS) requires milligram-scale quantities of purified protein. A similar analytical challenge arises in cosmochemistry. Amino acid $\delta^{15}N$ values provide critical constraints on the origin and evolution of extraterrestrial organic matter [3,4], yet measurements remain scarce due to low compound abundances.

Here we present methodological advances based on GC-Orbitrap-based Fourier-transform mass spectrometry that overcome these limitations. Our approach enables isotopic measurements at natural abundance from sample sizes as small as ~1 ng. A recent study using similar methods reports N isotope data for amino acids in the Murchison meteorite and samples of the asteroid Bennu [5,6], with precisions of ~3–32‰. The method we present routinely reaches sub-per mil precisions, more appropriate for palaeobiological and ecological applications. We report the first amino acid $\delta^{15}N$ measurements of modern dental enamel, using GC-IRMS as a validation benchmark. These results establish a foundation for extending amino acid–specific nitrogen isotope analyses to fossil enamel, opening new opportunities to reconstruct diets and trophic ecology across deep time. More broadly, this framework enables high precision $\delta^{15}N$ measurements of nitrogen-bearing organic compounds in meteorites, returned space samples, and other abundance-limited materials.

References: [1] Chikaraishi et al., (2009) Limnol. Oceanogr. Methods. 740–750; [2] Hkouchi, (2023) Proc. Jpn. Acad., Ser. B. 99, 131–154; [3] Chan, et al. (2016)., Earth, Planets Sp. 68, 7; [4] Alexander, et al. (2007), Geochim. Cosmochim. Acta. 71, 4380–4403; [5] McIntosh, et al. (2025), Rapid Commun. Mass Spectrum. 39, e2517723123; [6] Baczynski, et al. (2026), PNAS, 1–8.

Walters et al., "Development of a Low-Mass, High-Precision Multi-Isotope Method for Sulfate ($\delta^{34}S$, $\Delta^{33}S$, $\delta^{18}O$, $\Delta^{17}O$) Using ESI-Orbitrap Mass Spectrometry"

Multiple sulfur and oxygen isotope measurements of sulfate (SO_4^{2-}) are powerful tools for constraining atmospheric oxidation chemistry and biogeochemical sulfur cycling. However, conventional isotope ratio mass spectrometry (IRMS) techniques typically require several micromoles of SO_4^{2-} and involve extensive chemical conversion, limiting applications in low-mass environmental systems. Here we present a refined electrospray ionization–Orbitrap (ESI-Orbitrap) method for direct isotopologue analysis of intact SO_4^{2-}, achieving precisions of ±0.2‰ (δ^{34}S), ±0.3‰ (Δ^{33}S), ±0.3‰ (δ^{18}O), and ±0.4‰ (Δ^{17}O). Method development required systematic evaluation of solvent composition, SO_4^{2-}concentration effects, dehydration and resuspension protocols, fragmentation behavior, and Orbitrap resolution settings. Optimized sample preparation includes offline cation exchange resin pretreatment, chromatographic sulfate isolation and collection, quantitative water removal, and reconstitution in 90:10 methanol:water to enhance ionization efficiency and minimize matrix effects. Using this workflow, we demonstrate reliable sulfate isotope analysis from as little as ~100 nmol of SO_4^{2-}, with analytical limits governed by purification efficiency rather than instrumental sensitivity. This represents an order-of-magnitude reduction in sample size relative to traditional IRMS approaches. We apply the method to diverse environmental systems, including fine-mode aerosol collected by aerosol impactors, atmospheric oxidation experiments of reduced sulfur gases, and wastewater treatment samples. The ability to obtain coupled δ^{34}S–Δ^{33}S–δ^{18}O–Δ^{17}O measurements from low-mass sulfate opens new opportunities for investigating sulfur cycling in atmospheres, hydrospheres, and engineered systems where material is limited.

Poster Presentations

None noted.